

```
mysql> INSERT INTO categories (name, description) VALUES
-> ('Electronics', 'Gadgets, devices and accessories'),
-> ('Clothing', 'Apparel for men, women and children'),
-> ('Books', 'Fiction, non-fiction and educational'),
-> ('Home & Kitchen', 'Appliances and cookware'),
-> ('Sports', 'Fitness and outdoor equipment');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO customers (full_name, email, phone) VALUES
-> ('Alice Johnson', 'alice@example.com', '9876543210'),
-> ('Bob Smith', 'bob@example.com', '9123456789'),
-> ('Carol White', 'carol@example.com', '9000011122'),
-> ('David Kumar', 'david@example.com', NULL),
-> ('Eva Patel', 'eva@example.com', '9988776655');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> INSERT INTO products (category_id, name, price, stock_qty) VALUES
-> (1, 'Wireless Headphones', 2499.00, 50),
-> (1, 'USB-C Hub', 799.00, 120),
-> (1, 'Mechanical Keyboard', 3299.00, 30),
-> (2, 'Cotton T-Shirt', 499.00, 200),
-> (2, 'Denim Jeans', 1299.00, 80),
-> (3, 'SQL Mastery Book', 599.00, 60),
-> (3, 'Data Structures Guide', 749.00, 45),
-> (4, 'Stainless Steel Kettle', 999.00, 35),
-> (5, 'Yoga Mat', 849.00, 90),
-> (5, 'Resistance Bands Set', 349.00, 150);
Query OK, 10 rows affected (0.00 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO orders (customer_id, status, total_amount) VALUES
-> (1, 'delivered', 3298.00), -- Alice: headphones + hub
-> (2, 'shipped', 1299.00), -- Bob: jeans
-> (3, 'confirmed', 1348.00), -- Carol: books
-> (4, 'pending', 3299.00), -- David: keyboard
-> (1, 'confirmed', 849.00), -- Alice: yoga mat (second order)
-> (5, 'delivered', 1848.00); -- Eva: kettle + resistance bands
Query OK, 6 rows affected (0.01 sec)
Records: 6 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO order_items (order_id, product_id, quantity, unit_price) VALUES
-> (1, 1, 1, 2499.00), -- Alice → Headphones
-> (1, 2, 1, 799.00), -- Alice → USB-C Hub
-> (2, 5, 1, 1299.00), -- Bob → Jeans
-> (3, 6, 1, 599.00), -- Carol → SQL Book
-> (3, 7, 1, 749.00), -- Carol → DS Guide
-> (4, 3, 1, 3299.00), -- David → Keyboard
-> (5, 9, 1, 849.00), -- Alice → Yoga Mat
-> (6, 8, 1, 999.00), -- Eva → Kettle
-> (6, 10, 2, 349.00); -- Eva → Resistance Bands x2
Query OK, 9 rows affected (0.01 sec)
Records: 9 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> SELECT oi.order_id, p.name AS product, oi.quantity, oi.unit_price
-> FROM    order_items oi
-> JOIN    products p ON oi.product_id = p.product_id
-> ORDER  BY oi.order_id;
```

order_id	product	quantity	unit_price
1	Wireless Headphones	1	2499.00
1	USB-C Hub	1	799.00
2	Denim Jeans	1	1299.00
3	SQL Mastery Book	1	599.00
3	Data Structures Guide	1	749.00
4	Mechanical Keyboard	1	3299.00
5	Yoga Mat	1	849.00
6	Stainless Steel Kettle	1	999.00
6	Resistance Bands Set	2	349.00

9 rows in set (0.00 sec)

```
mysql> SELECT item_id, order_id, product_id, quantity, unit_price
-> FROM    order_items;
```

item_id	order_id	product_id	quantity	unit_price
1	1	1	1	2499.00
2	1	2	1	799.00
3	2	5	1	1299.00
4	3	6	1	599.00
5	3	7	1	749.00
6	4	3	1	3299.00
7	5	9	1	849.00
8	6	8	1	999.00
9	6	10	2	349.00

9 rows in set (0.00 sec)

```
mysql>
mysql> SELECT
-> o.order_id,
-> c.full_name      AS customer_name,
-> c.email          AS customer_email,
-> o.status,
-> o.total_amount,
-> o.ordered_at
-> FROM    orders o
-> JOIN    customers c ON o.customer_id = c.customer_id
-> ORDER  BY o.order_id;
```

order_id	customer_name	customer_email	status	total_amount	ordered_at
1	Alice Johnson	alice@example.com	delivered	3298.00	2026-06-24 11:15:28
2	Bob Smith	bob@example.com	shipped	1299.00	2026-06-24 11:15:28
3	Carol White	carol@example.com	confirmed	1348.00	2026-06-24 11:15:28
4	David Kumar	david@example.com	pending	3299.00	2026-06-24 11:15:28
5	Alice Johnson	alice@example.com	confirmed	849.00	2026-06-24 11:15:28
6	Eva Patel	eva@example.com	delivered	1848.00	2026-06-24 11:15:28

6 rows in set (0.00 sec)

```
mysql> SELECT 'categories' AS table_name, COUNT(*) AS rows_count
-> FROM categories
->
-> UNION ALL
->
-> SELECT 'customers', COUNT(*)
-> FROM customers
->
-> UNION ALL
->
-> SELECT 'products', COUNT(*)
-> FROM products
->
-> UNION ALL
->
-> SELECT 'orders', COUNT(*)
-> FROM `orders`
->
-> UNION ALL
->
-> SELECT 'order_items', COUNT(*)
-> FROM order_items;
```

table_name	rows_count
categories	5
customers	5
products	10
orders	6
order_items	9

5 rows in set (0.00 sec)

```
mysql> SELECT
-> o.order_id,
-> c.full_name          AS customer,
-> p.name               AS product,
-> cat.name             AS category,
-> oi.quantity,
-> oi.unit_price,
-> (oi.quantity * oi.unit_price) AS line_total,
-> o.status
-> FROM orders o
-> JOIN customers c ON o.customer_id = c.customer_id
-> JOIN order_items oi ON o.order_id = oi.order_id
-> JOIN products p ON oi.product_id = p.product_id
-> JOIN categories cat ON p.category_id = cat.category_id
-> ORDER BY o.order_id, p.name;
```

order_id	customer	product	category	quantity	unit_price	line_total	status
1	Alice Johnson	USB-C Hub	Electronics	1	799.00	799.00	delivered
1	Alice Johnson	Wireless Headphones	Electronics	1	2499.00	2499.00	delivered
2	Bob Smith	Denim Jeans	Clothing	1	1299.00	1299.00	shipped
3	Carol White	Data Structures Guide	Books	1	749.00	749.00	confirmed
3	Carol White	SQL Mastery Book	Books	1	599.00	599.00	confirmed
4	David Kumar	Mechanical Keyboard	Electronics	1	3299.00	3299.00	pending
5	Alice Johnson	Yoga Mat	Sports	1	849.00	849.00	confirmed
6	Eva Patel	Resistance Bands Set	Sports	2	349.00	698.00	delivered
6	Eva Patel	Stainless Steel Kettle	Home & Kitchen	1	999.00	999.00	delivered

9 rows in set (0.00 sec)

```
mysql> SELECT product_id, name, stock_qty
-> FROM products
-> WHERE stock_qty < 40
-> ORDER BY stock_qty ASC;
```

product_id	name	stock_qty
3	Mechanical Keyboard	30
8	Stainless Steel Kettle	35

2 rows in set (0.00 sec)

```
mysql> SELECT
-> c.customer_id,
-> c.full_name,
-> COUNT(o.order_id) AS total_orders,
-> SUM(o.total_amount) AS total_spent
-> FROM customers c
-> LEFT JOIN orders o ON c.customer_id = o.customer_id
-> GROUP BY c.customer_id, c.full_name
-> ORDER BY total_spent DESC;
```

customer_id	full_name	total_orders	total_spent
1	Alice Johnson	2	4147.00
4	David Kumar	1	3299.00
5	Eva Patel	1	1848.00
3	Carol White	1	1348.00
2	Bob Smith	1	1299.00

5 rows in set (0.00 sec)